BROOKBANKS

# 1335 Swindon Nev

Technical Note TN47- OVL/A420 Signalised Junction S278 Surface Water Design
24th January 2025

## 1 Executive Summary

**1.1** Swindon Eastern Villages Consortium have appointed Brookbanks to produce a surface water drainage design for a residential-led scheme development at South Marston and Rowborough to the east of Swindon, together with areas of mixed-use comprising community; employment; education; local retail uses; and formal and informal open space provision on land east of Swindon and north of A420.

**1.2** The scheme is to include an upgrade to the existing junction between the A420 and Old Vicarage Lane (OVL) to be adopted under a S278 agreement. The junction will require an approved drainage system to discharge Condition 27– Old Vicarage Lane/A420 (ref. S/OUT/13/1555).

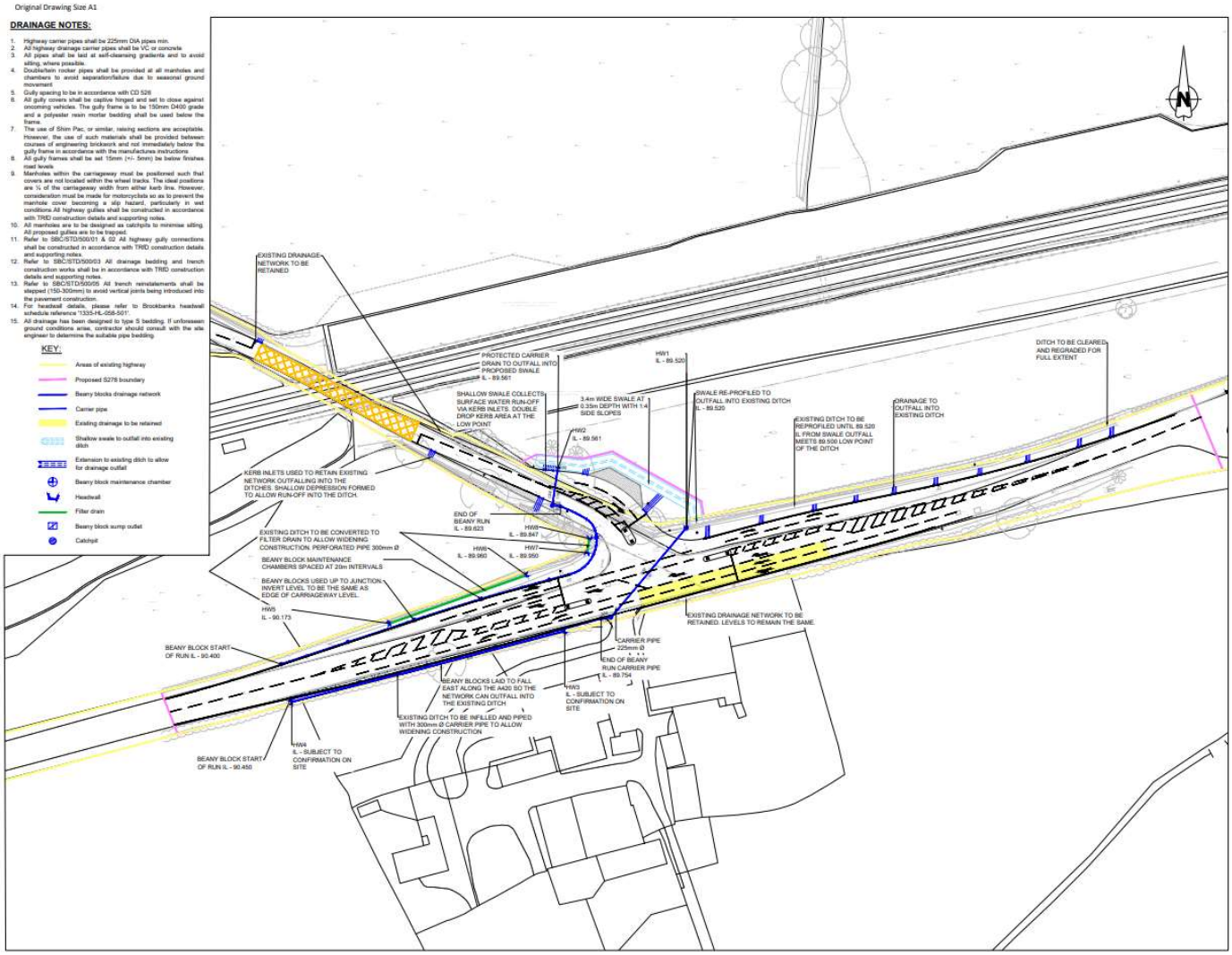
**1.3** We believe this can be achieved with the provision of additional gullies (in the form of beany blocks) and carrier drains along the A420 in combination with a swale and utilizing the existing ditch to attenuate and treat the surface water run off as set out in this note and on drainage plan ref **1335-HL-058-500L Drainage Layout**. The proposal accords with the approved Flood Risk Assessment 1335 FRA01 Rv3 and the reports prepared to discharge Conditions 40 & 41 relating to site wide drainage and flood mitigation.

# 2 Proposed Junction

**2.1** The proposed layout consists of

- An upgraded signalised junction on the A420
- A maintenance layby
- A swale outfalling into the existing ditch
- Kerb inlets on the eastern side of the junction

Shown in the extract below.




**Proposed drainage layout (Extract Brookbanks Drainage GA ref 1335 -HL-58-500L)**

# 3 Surface Water Run Off

### Existing Highway Drainage System

**3.1** A topographical survey has been undertaken to verify the extent of the existing highway drainage system draining the OVL/A420. Brookbanks have visited site for a walkover and observed the existing highway drainage system to consist of a combination of over the edge drainage and gully inlets to an existing ditch on either side of the A420.

**3.2** The proposal will be to retain and improve the existing drainage system where the alignment remains unchanged.

### Proposed Highway SuDS

**3.3** The S278 works will result in an increase of surface water runoff. In line with approved FRA and SWDS a new highway SuDS system is proposed to treat and attenuate runoff as illustrated in **Drawing 1335-HL-058-500 Drainage Layout**.

**3.4** The proposal will include

- A swale to intercept the runoff from Old Vicarage Lane, also providing an outfall for the northern A420 beany block network
- Kerb inlets along the eastern side of the OVL to outfall into the existing ditch
- Beany block network along the west of the A420 before outfalling into the proposed swale/existing ditch
- Outfall to the existing ditchcourse

**A typical swale arrangement will be 3.4m wide approx. 0.5m deep with 1 in 3 side slopes.**

The primary aims of the swale is as follows:

- Final flow and water quality conditioning
- Provide landscaping, amenity and ecological benefits

### Outfall

**3.5** Flows will be restricted by a control device and will outfall to the existing ditch on the northside of the A420 Clearance work will be undertaken to the existing ditch course to ensure flow is maintained.

### Supporting measurements and calculations

**3.6** The proposed highway drainage arrangements, including the incorporation of a swale and improved drainage networks on the A420/OVL, have resulted in no increase to the flow rates to the existing highway system or surrounding drainage network up to and including the 1 in 100-year storm event plus 40% climate change.

# 4 Maintenance Plan

### Highway Drains

**4.1** Once constructed and when the adoption process has been completed, Swindon Borough Council will be responsible for maintaining the highway drainage networks and ensuring their continued operation. Consequently drains, manhole and headwalls will be maintainable by the SBC.

### SuDS

**4.2** The swale is to be maintained by Swindon Borough Council in accordance with a Maintenance & Management Plan to be approved by the LPA/LLFA. Detailed plans and design drawings indicating features and areas which will be maintained by Swindon Borough Council will be provided in the Maintenance Plan.

**4.3** The development layout provides adequate access and working areas to undertake the above maintenance procedures.

**4.4** The Council maintenance team will maintain the basin in accordance with best practice based on a typical maintenance regime shown in the tables below:

| Maintenance Schedule | Required Action | Typical Frequency |
|---|---|---|
| Regular Maintenance | Remove litter (including leaf litter) and debris | Monthly or as required |
| | Cut grass for spillways and access routes | Monthly during growing season or as required |
| | Manage other vegetation and remove nuisance plants | Monthly at start, then as required |
| | Inspect inlets, outlets and overflows for blockages, and clear if required. | Monthly |
| | Inspect banksides, structures, pipework etc for evidence of physical damage | Monthly or when required |

| | | |
|---|---|---|
| | Inspect inlets and facility surface for silt accumulation. Establish appropriate silt removal frequencies | Monthly for 12 months, then annually or as required |
| | Inspect inlets and facility surface for silt accumulation, and establish appropriate silt removal frequencies | Six Monthly |
| | Check any penstocks and other mechanical devices | Annually |
| | Tidy all dead growth before start of growing season | Annually |
| | Remove sediment from inlets, outlet and forebay | Annually or as required |
| | Manage wetland plants in outlet pool- where provided | Annually |
| Occasional Maintenance | Reseed areas of poor vegetation growth. | As required |
| | Prune and trim any trees and remove cuttings | Every 2 years or as required |
| | Remove sediment from inlets, outlet and forebay and main basin | Every 5 years or as required |
| Remedial Actions | Repair erosion or other damage by re-turfing or reseeding | As required |
| | Re-alignment of riprap | As required |
| | Repair/ rehabilitate inlets, outlets and overflows | As required |
| | Relevel uneven surfaces and reinstate design levels. | As required |

### Exceedance Flows

**4.5** The wide verge and swale will allow a route for exceedance flows ensuring any flooding in extreme events is diverted away from habitable dwellings.

# 5 Next Steps and Recommendations

**5.1** The proposed swale system is designed in accordance with the approved FRA to allow deliver of the proposed S278 signalised junction.

**5.2** The developer to ensure that the contractor constructs and maintains the drainage system until the system is adopted by Swindon Borough Council.